

RAHUL AJAYKUMAR MADHAWANI

Software Engineer — Full-Stack & AI Systems

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student (CGPA 8.48/10) with hands-on experience building full-stack and multi-agent AI systems in React Native, Python, FastAPI, and Google ADK/Gemini. Built an automated NLP-driven verification pipeline that cut manual review effort by ~70% during an industry project, and contributed a merged pull request enabling WebAssembly support in an open-source Rust AI-memory engine. Seeking full-time Software Engineer roles in full-stack development or AI/ML systems.

TECHNICAL SKILLS

Languages: C++, Java, Python, TypeScript, C

Frameworks & Libraries: React.js, React Native, Node.js, Express.js, FastAPI, Google ADK

Databases: PostgreSQL, Redis, Kuzu (graph DB), LanceDB (vector DB)

Cloud & DevOps: AWS, Docker, Firebase, Appwrite, GitHub Actions

Testing & Tools: Pytest, REST API Testing & Automation, Git/GitHub, NLP

Fundamentals: Object-Oriented Programming, Operating Systems, Computer Networks, Linux (Basic)

PROFESSIONAL EXPERIENCE

Passion Infotech Pvt. Ltd. — Industry Project Intern

Mar 2025 – Apr 2025

Project: Impact Analysis of Funded Incubatees

- Engineered an automated verification system using a Python web-scraping pipeline (BeautifulSoup) to collect digital footprints of government-funded entrepreneurs from LinkedIn and search engines.
- Applied NLP techniques to cross-reference collected data against funding claims, reducing manual verification effort by approximately 70% and accelerating the validation process.
- Wrote Pytest-based unit and integration tests for the scraping and NLP modules, improving pipeline reliability and catching data-validation issues before deployment.

Tech: React.js, Node.js, Express.js, Python, Pytest, BeautifulSoup, NLP

PROJECTS

Cognivest — Per-Company Financial Intelligence Platform [\[GitHub\]](#) | *FastAPI, Next.js 14, Cognee, Kuzu, LanceDB, Gemini, PostgreSQL, Docker*

- Built a knowledge-graph platform fusing price action with news/web narrative per stock ticker to answer sourced, cited questions about price moves.
- Designed a four-stage AI memory lifecycle (Remember → Recall → Improve → Forget) with per-ticker dataset isolation, running end-to-end on free-tier infrastructure at zero cost.

FIFA Rapid Agent 2026 — AI Football Companion App [\[GitHub\]](#) | *React Native (Expo), Firebase, Google ADK, Gemini 2.0 Flash, FastAPI*

- Built a real-time football companion app with live match data, an SVG formation chart, a draggable live-score capsule, and Firebase (Email + Google OAuth) authentication.
- Architected a multi-agent backend (root agent routing to Match, Stats, and Tournament sub-agents) with screen-context injection so the assistant answers questions about the user's current view.

SmartLedger — Voice-Driven Billing Assistant [\[GitHub\]](#) | *React Native, Appwrite, Whisper, JWT Auth*

- Built a cloud-based billing and inventory app for retailers with voice-driven billing, multilingual speech recognition, and real-time inventory sync via Appwrite.
- Designed REST-based mobile-to-backend communication with JWT authentication; tested billing/inventory endpoints for correctness and data integrity.

NarratorApp — Real-Time Environment Narrator for the Visually Impaired [\[GitHub\]](#) | *Kotlin, CameraX, TensorFlow Lite, Google ML Kit, TTS*

- Built an Android app performing on-device object detection and text recognition, converting visual context into real-time audible narration for visually impaired users.

EDUCATION

Vishwakarma Institute of Technology, Pune — B.Tech, Computer Engineering (CGPA: 8.48/10)

2022 – 2026 (Expected)

Late Jyoti Janolkar College, Akola — HSC (81.8%)

2021 – 2022

CERTIFICATIONS & ACHIEVEMENTS

- Cognee (topoteretes/cognee-rs) — Open-Source Contribution, Rust AI Memory Engine** [\[GitHub\]](#) | *Rust, WebAssembly, Tokio*
Landed a merged PR making three core crates compile and run on wasm32-unknown-unknown, verified end-to-end in headless Chrome; resolved WASM incompatibilities in randomness, async runtime, and time handling behind target-specific guards.
- NVIDIA Deep Learning Institute — Fundamentals of Deep Learning | Coursera — AWS Cloud Technology Consultant
- Hackathon Participant — Adobe India, CodeClash, Flipkart Grid, Tata Imagination Challenge